

Graph Neural Networks and Graph Foundation Models for Scientific Discovery

Abstract

The integration of Graph Neural Networks (GNNs) and Graph Foundation Models (GFM) into scientific discovery represents a paradigm shift from task-specific modeling to universal, transferable representation learning. This survey synthesizes recent advancements across 300 academic studies to provide a comprehensive analysis of how graph-based architectures are reshaping domains ranging from molecular drug discovery and materials science to computational physics and bibliometric analysis. We categorize the field into distinct methodological families, including structure-centric pretraining strategies, language-model-driven graph textualization, geometric and equivariant architectures, and retrieval-augmented reasoning systems. Our analysis reveals that while early GNNs were limited by domain specificity and data scarcity, modern GFMs leverage self-supervised objectives, synthetic priors, and cross-domain transfer mechanisms to achieve state-of-the-art performance in zero-shot and few-shot settings. We critically examine the emergence of scaling laws in graph learning, the role of mechanistic simulation as supervision, and the deployment of multi-agent systems for autonomous hypothesis generation. Furthermore, we identify persistent challenges regarding out-of-distribution generalization, computational scalability, and the interpretability of black-box models in high-stakes scientific applications. By mapping the trajectory from specialized encoders to general-purpose graph intelligence, this work outlines a roadmap for the next generation of AI-driven scientific infrastructure.

1. Introduction

Scientific discovery has historically been driven by the interplay of theoretical formulation, experimental observation, and computational simulation. In recent years, machine learning has emerged as a fourth pillar, with Graph Neural Networks (GNNs) playing a central role due to their natural affinity for representing relational data inherent in scientific systems [1]. From the atomic bonds in molecules [2] and the crystal lattices in materials [3] to the citation networks of academic literature [4] and the interaction topologies in biological pathways [5], graph structures provide a universal language for encoding complex dependencies. However, the traditional paradigm of training specialized GNNs for isolated tasks faces significant bottlenecks: data scarcity in high-cost experimental domains, the inability to generalize across disparate graph distributions, and the lack of emergent reasoning capabilities observed in Large Language Models (LLMs) [6].

The concept of Graph Foundation Models (GFMs) has arisen to address these limitations, aiming to construct unified models pretrained on broad corpora of graph data that can be adapted to diverse downstream tasks with minimal fine-tuning [7]. Unlike their language counterparts, GFMs must contend with the non-Euclidean nature of graph data, the heterogeneity of node features, and the absence of a canonical tokenization scheme [6]. Recent literature suggests that successful GFMs require redesigned backbones and unified task formulations to achieve “emergence” and “homogenization” across domains [6]. Approaches range from purely structural pretraining using contrastive and generative objectives [8] to hybrid architectures that leverage LLMs for graph textualization and instruction tuning [9].

This survey provides a systematic review of the current landscape of GNNs and GFMs in scientific discovery. We synthesize evidence from 300 relevant papers to construct a detailed taxonomy of methodological innovations. We explore how self-supervised learning strategies are evolving from

simple masking tasks to complex mechanistic simulations [10], how geometric deep learning is incorporating Riemannian manifolds and equivariance constraints to respect physical laws [11][12], and how retrieval-augmented generation is being adapted for graph-structured knowledge [13]. Furthermore, we examine the burgeoning field of autonomous scientific agents that utilize graph reasoning for hypothesis generation and experimental design [14][15].

The motivation for this survey is twofold. First, the rapid proliferation of GFM architectures necessitates a coherent framework to compare their underlying assumptions, particularly regarding transferability and scaling behaviors [16]. Second, the application of these models to scientific discovery introduces unique requirements for uncertainty quantification, causal inference, and physical consistency that differ from standard benchmark performance [17]. By analyzing the convergence of graph learning, foundation models, and scientific methodology, we aim to identify the critical frontiers that will define the next decade of AI-assisted science.

2. Method

The methodological landscape of Graph Neural Networks and Graph Foundation Models for scientific discovery is vast and rapidly evolving. To provide a structured analysis, we organize the field into six primary methodological systems: (1) Self-Supervised Pretraining and Transfer Learning, (2) Language-Model-Driven Graph Foundation Models, (3) Geometric and Equivariant Architectures, (4) Retrieval-Augmented and Tool-Using Graph Systems, (5) Generative Models and Autonomous Discovery Agents, and (6) Benchmarking and Evaluation Frameworks. Each section synthesizes multiple related papers to highlight technical distinctions, innovations, and empirical findings.

2.1 Self-Supervised Pretraining and Transfer Learning Strategies

The cornerstone of Graph Foundation Models is the ability to learn transferable representations from unlabeled or weakly labeled data. This section examines the evolution of pretraining objectives, the debate over scaling laws, and the mechanisms for cross-domain transfer.

2.1.1 Contrastive, Generative, and Predictive Objectives Self-supervised learning (SSL) on graphs is broadly categorized into contrastive, generative, and predictive paradigms [18]. Contrastive methods maximize mutual information between different views of the same graph. For instance, GraphCL and its variants employ data augmentation strategies such as node dropping and edge perturbation to create positive pairs [19]. However, recent studies indicate that sophisticated masking distributions often yield negligible benefits over uniform sampling due to limited gains in mutual information with downstream tasks [20]. Generative approaches, such as GraphMAE, focus on reconstructing masked node features or graph structures. GraphMAE has been shown to excel in node clustering tasks but may struggle with link prediction due to its feature-only reconstruction objectives [19].

Predictive SSL involves solving pretext tasks such as link prediction or motif classification. A notable innovation is the use of “task-trees” to unify node, edge, and graph-level tasks into a single tree structure, enabling cross-task generalization [21]. The GIT (Graph Generality Identifier on Task-Trees) framework constructs these trees by merging computation trees of task-relevant nodes, employing a dual-view reconstruction loss to pretrain message-passing GNNs [21]. Similarly, the GraphProp method utilizes regression on normalized graph invariants (e.g., Lovasz number, Fiedler value) as a pretraining objective, forcing the model to learn domain-agnostic structural representations superior to text-based encodings [22].

Recent advancements have also explored multi-task learning to capture disjoint yet complementary knowledge. PARETOGNN employs the Multiple Gradient Descent Algorithm (MGDA) to optimize five distinct pretext tasks simultaneously, achieving Pareto optimality and enhancing task generalization [23]. Furthermore, the concept of “Graph Interplay” introduces stochastic inter-graph edges within training batches to facilitate direct information flow across disparate graphs, acting as an implicit regularizer that enhances manifold separation [24].

2.1.2 Scaling Laws and the Pretraining-Downstream Gap A critical question in the development of GFMs is whether neural scaling laws, observed in NLP and vision, hold for graph data. Empirical benchmarks suggest a complex picture. While model width and label diversity drive downstream performance in certain regimes, scaling trends often plateau or degrade for depth beyond 8-16 layers due to oversmoothing [25]. A systematic benchmark evaluating InfoGraph, GraphCL, and GraphMAE found that downstream performance frequently fails to scale with data or model size despite decreasing SSL loss, indicating a significant gap between pretraining objectives and downstream utility [8].

This disconnect is further evidenced by the finding that pretrained embeddings enhance generalization only when sufficient downstream data exists; in low-data regimes, multi-dataset pretraining mitigates domain gaps but does not guarantee superior performance over non-pretrained baselines [26]. The “GraphFM” benchmark highlights that generative models excel in node clustering but fail in link prediction, and no single early stopping criterion optimizes multi-task generalization [19]. Consequently, the field is moving towards more nuanced scaling strategies, such as the “Universal Self-Supervised Learning” (U-SSL) framework, which employs graph-specific encoders to homogenize disparate feature dimensions before processing them with a shared universal module [27].

2.1.3 Cross-Domain Transfer and Structural Prompts Transfer learning in GFMs often relies on structural prompts or feature-agnostic encodings to bridge domain gaps. The “Feature-Structuralization” method transforms categorical node features into auxiliary feature-nodes, discarding explicit feature vectors to retain only topological marks, thereby enabling feature-agnostic processing [26]. Similarly, structural prompting converts topological statistics (degree, centrality, k-core) into natural language profiles, which are encoded via MiniLM and aligned with raw features through contrastive learning [28]. This approach has demonstrated a +21.8% improvement over supervised baselines on protein-protein interaction datasets [28].

Another promising direction is the use of “computation trees” as transferable units. The GFT (Graph Foundation model with transferable Tree vocabulary) redefines transferable patterns as computation trees derived from message-passing, utilizing vector quantization to generate a discrete tree vocabulary [29]. This method exceeds baseline performance by an average of over 6% across cross-domain datasets [29]. Additionally, the “Graph Vocabulary” perspective advocates for constructing basic transferable units grounded in network analysis and stability principles, separating vocabularies for conflicting graph patterns to improve robustness [30].

2.2 Language-Model-Driven Graph Foundation Models

The convergence of LLMs and GNNs has given rise to a new class of models that leverage the semantic reasoning capabilities of language models for graph tasks. This section explores strategies for graph textualization, instruction tuning, and the integration of LLMs as graph encoders.

2.2.1 Graph Textualization and Instruction Tuning A fundamental challenge in LLM-based graph learning is converting non-Euclidean structures into sequences compatible with transformer architectures. LangGFM addresses this by converting graph structures into natural language using standard exchange formats like GraphML, GML, and JSON [9]. By employing joint instruction tuning on Llama-3-8B-Instruct, LangGFM treats format diversity as a form of language-space data augmentation, achieving a 5.5% accuracy improvement over single-format baselines on shortest path tasks [9].

Similarly, the “One for All” (OFA) framework converts heterogeneous graph data into Text-Attributed Graphs (TAGs) using LLM encoders to unify feature spaces [31]. OFA standardizes node, link, and graph tasks via “Nodes-of-Interest” (NOI) subgraphs connected to a task-specific prompt node, enabling zero-shot and few-shot learning across domains [31]. The “UniGraph” model further advances this by employing a cascaded LM-GNN encoder with Graph Siamese Masked Autoencoders for self-supervised pretraining on TAGs, demonstrating that text serves as a universal medium to align feature spaces across diverse graph domains [32].

2.2.2 LLMs as Graph Encoders and Reasoners Beyond textualization, recent works explore using LLMs directly as graph encoders or reasoners. The “PromptGFM” method replicates the GNN message-passing workflow entirely in text space using iterative LLM prompting, establishing a universal language-based graph vocabulary for node representation [33]. This approach captures structural dependencies without numerical embeddings, outperforming traditional GCNs and GATs on node classification tasks [33].

In the realm of scientific discovery, LLMs are increasingly used to extract and reason over graph-structured knowledge. The “DrugChat” system bridges GNNs and LLMs via a linear adaptor, enabling interactive molecular conversation and natural language querying of drug properties [34]. For citation graphs, “LitFM” integrates a neighbor-aware graph retriever with a knowledge-infused LLM, utilizing pseudo-query embeddings to handle sparse contexts and improving link prediction accuracy by 8.0% over GAT baselines [35]. Furthermore, the “Intern-Atlas” project constructs a massive methodological evolution graph from over one million AI papers, using LLMs to classify causal relations between methods and enabling deterministic reasoning over scientific progress [36].

2.2.3 Hybrid Architectures and Modality Alignment Hybrid architectures that tightly couple GNNs and LLMs offer a balance between structural fidelity and semantic richness. The “GraphFormers” architecture nests layerwise GNN components alongside Transformer blocks, allowing neighborhood context to iteratively refine token representations [37]. This fusion has shown relative improvements of 2.9% in precision for link prediction on bibliographic networks [37].

For text-attributed graphs, “GraphCLIP” generates large-scale graph-summary pairs using LLMs and aligns modalities via a shift-robust contrastive loss, achieving a 15.26% accuracy gain in zero-shot settings [38]. The “TAGA” framework synergizes graph and text mutual transformations by constructing complementary “Text-of-Graph” and “Graph-of-Text” views, utilizing hierarchical document layouts to preserve topology during textualization [39]. These methods underscore the importance of aligning structural and semantic modalities to leverage the strengths of both architectures.

2.3 Geometric and Equivariant Architectures

Scientific data often resides on non-Euclidean manifolds or possesses inherent symmetries (e.g., rotation, translation) that must be respected by the model. This section reviews advances in Riemannian graph learning, equivariant networks, and geometric foundation models.

2.3.1 Riemannian Geometry and Manifold Learning Traditional GNNs operate in Euclidean space, which may be ill-suited for hierarchical or cyclic graph structures. Riemannian graph learning addresses this by endowing GNNs with intrinsic manifold properties. The “RiemannGL” taxonomy categorizes methods by manifold type, neural architecture, and learning paradigm, demonstrating that intrinsic properties like curvature provide superior inductive biases for specific topologies [11].

Building on this, “RiemannGFM” constructs a product bundle combining hyperbolic and hyperspherical tangent bundles to model tree and cycle substructures respectively [40]. It introduces a vocabulary learning module that utilizes cross-geometry attention to update manifold coordinates, outperforming open graph models on node classification tasks [40]. Another approach, “GAUGE” (Graph Universal Pretraining with Intrinsic Geometry), constructs a Neural Vector Bundle over the graph manifold, employing parametric local trivialization and a Dirichlet loss to measure behavioral smoothness [41]. GAUGE posits that transferable substructures correspond to regions of geometric flatness, achieving state-of-the-art results in few-shot cross-domain transfer [41].

2.3.2 Equivariance and Symmetry Constraints In physics and chemistry, models must often be equivariant to transformations such as rotation and translation. “Triple-Symmetry Graph Neural Networks” (TS-GNNs) enforce node permutation-equivariance, label permutation-equivariance, and feature permutation-invariance simultaneously, enabling zero-shot transfer across domains with disjoint label spaces [12]. Evaluation on 28 unseen datasets shows that TS-GNNs exceed baselines like GraphAny when trained with these symmetry constraints [12].

For 3D molecular graphs, “3D-EMGP” utilizes an $E(3)$ -equivariant GNN backbone to predict atomic forces as negative gradients of scalar energy, employing a Riemann-Gaussian noise distribution for denoising [42]. This physics-inspired pretraining aligns the model with physical laws, outperforming 2D baselines on force prediction tasks [42]. Similarly, “Equivariant Graph Attention Networks” (EQGAT) process point clouds using separate scalar and vector feature streams, computing content-and-spatial dependent filters to preserve geometric fidelity with fewer parameters than spherical harmonic-based methods [43].

2.3.3 Geometric Bases and Metric Measure Spaces A novel perspective models graphs as metric measure spaces aligned to learnable geometric bases. The “Structure-Centric Graph Foundation Model” (SCGFM) computes structural coordinates representing transport discrepancies to these bases via Sliced Gromov-Wasserstein distance [44]. This approach creates a shared structural coordinate system agnostic to node features, demonstrating substantial robustness under large structural shifts where baselines degrade to chance-level accuracy [44]. Similarly, “SPG” (Spectral Parsing and Prototype-Guided Spatial Propagation) applies learnable Chebyshev filters to decompose node features into frequency-aware spectral responses, constructing a shared Gromov-Wasserstein prototype geometry to distill transferable knowledge [45].

2.4 Retrieval-Augmented and Tool-Using Graph Systems

As scientific problems grow in complexity, single-model approaches are often insufficient. Retrieval-Augmented Generation (RAG) and tool-learning frameworks enable models to access external knowledge and execute complex reasoning chains.

2.4.1 Retrieval-Augmented Graph Learning Retrieval mechanisms allow GNNs to access relevant subgraphs or knowledge bases during inference. “RAGRAP” constructs a toy graph vector library via inverse importance sampling, retrieving top-K toy graphs to enrich the learning context beyond simple input-output mapping [13]. This dual-propagation mechanism improves accuracy by 5.19% on static graph tasks [13].

In the context of scientific literature, “LitFM” employs a neighbor-aware graph retriever that uses self-supervised contrastive learning to reconstruct citation connections, significantly improving precision over text-only RAG [35]. For materials science, a unified knowledge injection framework integrates a 3.5B-token combustion knowledge base with RAG, though it notes that naive RAG hits an accuracy ceiling of ~60% due to retrieval misses, necessitating knowledge-graph-enhanced approaches [46].

2.4.2 Tool Learning and API Invocation The integration of tools and APIs enables GFMs to perform actions beyond passive prediction. “ArtifactLinker” models the HuggingFace ecosystem as a heterogeneous graph, employing a dual-head GNN to predict link compatibility and conditional performance scores, effectively acting as a tool for automatic state-of-the-art discovery [47]. In laboratory automation, foundation models are being deployed to control robotic hardware via natural language, bypassing the need for uniform modularization [48]. The “AI Fluid Scientist” framework integrates hypothesis, experiment, and analysis agents that iteratively design experiments and control robotic water tunnels, discovering non-intuitive physical phenomena like spacing-independent anti-resonance frequencies [15].

2.4.3 Multi-Agent Systems for Scientific Discovery Multi-agent systems leverage the collective reasoning of multiple LLMs or specialized modules to solve complex scientific problems. “AgenticSciML” coordinates over ten specialized agents (Proposer, Critic, Engineer, etc.) in an evolutionary tree search, engaging in structured debates to refine solution proposals for partial differential equations [49]. This collaborative approach achieved an 11,169x improvement in error metrics over single-agent baselines [49].

Similarly, “RESEARCHTOWN” simulates a research community as an agent-data graph, where researcher agents and paper data nodes interact via text-space message passing to generate interdisciplinary ideas [50]. In the realm of causal discovery, “Auto-Bench” implements an autonomous iterative cycle where LLMs hypothesize adjacency matrices and propose interventions to an oracle, refining causal graph estimates until convergence [51]. These systems demonstrate that multi-agent collaboration can replicate real-world correlation patterns and accelerate the discovery process.

2.5 Generative Models and Autonomous Discovery

Generative models are pivotal for inverse design and hypothesis generation in scientific domains. This section covers graph generation, symbolic regression, and simulation-guided discovery.

2.5.1 Graph Generation and Inverse Design Generative models for graphs aim to create novel structures with desired properties. “AUTOGRAPH” transforms graphs into token sequences via Segmented Eulerian Neighborhood Trails (SENT), enabling autoregressive generation with linear complexity and 100x faster inference compared to diffusion-based models [52]. For molecular design, “MG2N2” executes sequential graph construction via independent GNN modules for node addition, edge classification, and parallel connection prediction, improving pattern generalization compared to sequence-only models [53].

In materials science, “DAO” (Siamese Foundation Models) employs a two-stage diffusion pretraining strategy that leverages unstable crystals relaxed by learned energy gradients, expanding the learning distribution and improving match rates in crystal structure prediction [54]. The “GFlowNets” framework learns a stochastic policy to construct compositional objects sequentially, addressing mode collapse by sampling proportionally to reward rather than maximizing it, thus preserving diversity in generated candidates [55].

2.5.2 Symbolic Regression and Equation Discovery Discovering underlying physical laws from data is a key goal of scientific AI. “AI Feynman 2.0” utilizes gradient analysis to detect graph modularity in computational graphs, recursively decomposing functions to enable efficient symbolic regression [56]. “GN-SR” (Graph Network - Symbolic Regression) trains a Graph Network simulator to approximate acceleration fields, then isolates the learned edge functions to recover symbolic equations, reducing mass estimation error to 1.6% in orbital mechanics tasks [57].

Neural approaches like “NICE” (Neural Infalling Cloud Equations) embed MLPs within coupled ODEs to replace uncertain analytic terms, utilizing symbolic regression post-processing to recover dimensionally consistent equations [58]. Similarly, “Parsimonious Neural Networks” integrate evolutionary optimization to minimize a composite objective balancing error and complexity, forcing the network to discover fundamental physical laws and stable integration schemes [59].

2.5.3 Simulation as Supervision Mechanistic simulations provide a rich source of synthetic supervision for training scientific models. “Simulation-Grounded Neural Networks” (SGNNs) train on synthetic corpora generated by stochastic mechanistic simulators, pairing simulated trajectories with exact parameters to internalize scientific structure [10]. This approach yielded a 12x improvement over Chronos in COVID-19 forecasting [10]. In aquatic science, a Physics-Guided Foundation Model (PGFM) pretrains on simulated environmental systems generated by physics-based models, utilizing evolutionary strategies to optimize feature interactions [60]. These methods highlight the potential of “back-to-simulation” attribution for process-level mechanistic interpretability [10].

2.6 Benchmarking, Evaluation, and Safety

Robust evaluation is critical for assessing the readiness of GFMs for scientific deployment. This section discusses benchmarking frameworks, out-of-distribution generalization, and security concerns.

2.6.1 Comprehensive Benchmarks Several benchmarks have been developed to standardize GFM evaluation. “GraphFM” unifies eight state-of-the-art SSL methods under consistent hyperparameter search budgets, revealing that mini-batch training generally degrades performance on small datasets [19]. “TAGLAS” provides an atlas of text-attributed graph datasets, unifying 23+ heterogeneous graphs into a TAG format to enable single-model training across domains [61]. For materials science, “Mars-Bench” curates 20 existing datasets into unified formats for Mars imagery, supporting cross-domain partitioning by sensor type [62].

2.6.2 Out-of-Distribution Generalization OOD generalization remains a significant challenge. The “OOD-GFM” taxonomy distinguishes between homogeneous-task and heterogeneous-task GFMs, analyzing their handling of structural, domain, and task shifts [63]. Methods like “VecFormer” employ vector quantization to decouple attention complexity from graph size, improving OOD robustness by 8.67% on Cora [64]. “GOBLIN” executes inference-time operator basis discovery via Bayesian Optimization, adapting linear graph operators to specific tasks and outperforming fixed-architecture GFMs on heterogeneous benchmarks [65].

2.6.3 Security and Privacy As GFMs are deployed in sensitive scientific domains, security and privacy become paramount. “MEA-GFM” demonstrates that one-loss embedding regression suffices for zero-shot capability transfer, posing a risk of model extraction attacks [66]. In drug discovery, black-box membership inference attacks can successfully identify training molecules, particularly those from minority classes, raising concerns about data leakage [67]. Furthermore, “GFM-BA” introduces label-free trigger association via Farthest Point Sampling, enabling target-controlled backdoor attacks without knowledge of downstream labels [68]. These findings underscore the need for robust defense mechanisms and privacy-preserving training protocols.

3. Challenges and Outlook

Despite significant progress, the field of Graph Neural Networks and Graph Foundation Models for scientific discovery faces several critical challenges that must be addressed to realize their full potential.

Lack of Universal Scaling Laws: Unlike LLMs, GFMs have not yet demonstrated consistent scaling laws where performance monotonically improves with model size and data volume [8][16]. The pretraining-downstream task gap remains a persistent issue, where decreases in SSL loss do not correlate with improved downstream performance [8]. Future research must focus on designing pretraining objectives that better align with downstream scientific tasks, potentially through mechanistic simulation or task-tree unification [21][10].

Heterogeneity and Domain Shift: Scientific graphs are inherently heterogeneous, varying widely in size, topology, and feature semantics. Current GFMs often struggle with cross-domain transfer, particularly when source and target domains exhibit significant structural divergence [69]. While methods like structural prompting and geometric bases show promise [28][44], a unified framework for handling extreme heterogeneity remains elusive. The development of “graph vocabularies” that can encode invariance across diverse graphs is a promising direction but requires further theoretical grounding [30].

Computational Scalability: The quadratic complexity of attention mechanisms in Graph Transformers limits their applicability to large-scale scientific graphs, such as those found in genomics or climate modeling [70]. While sparse attention mechanisms and linear approximations like EXPHORMER and SGFormer offer relief [71][72], they often come at the cost of expressivity or require careful tuning of sparse patterns [71]. Efficient architectures that can scale to billion-edge graphs without sacrificing accuracy are urgently needed.

Interpretability and Trust: In high-stakes scientific applications, black-box models are insufficient. Scientists require interpretable models that can provide causal explanations and uncertainty estimates. While methods like FIGNN offer feature-specific interpretability [73] and evidential PINNs provide calibrated uncertainty [17], integrating these capabilities into large-scale GFMs remains challenging. The “Veridical Data Science” framework emphasizes the need for predictability,

computability, and stability in foundation model lifecycles, calling for FDA-like disclosures on data and algorithmic decisions [74].

Data Scarcity and Quality: Many scientific domains suffer from data scarcity, with high-quality labeled data being expensive or impossible to obtain. While self-supervised learning and synthetic data generation offer partial solutions [75][10], the reliance on heuristic augmentations or imperfect simulators can introduce biases. The development of robust methods for learning from noisy, incomplete, or multi-modal scientific data is a critical area for future work [76].

Safety and Security: As GFMs become integral to scientific infrastructure, they become targets for adversarial attacks. The vulnerability of GFMs to model extraction, backdoor attacks, and membership inference poses significant risks [66][68][67]. Developing robust defense mechanisms and secure training protocols is essential for the safe deployment of these models in sensitive domains like drug discovery and national security.

4. Conclusion

The integration of Graph Neural Networks and Graph Foundation Models into scientific discovery marks a transformative era in computational science. By leveraging self-supervised pretraining, geometric deep learning, and language-model integration, these models are overcoming the limitations of data scarcity and domain specificity that have long plagued scientific AI. From the atomic scale of molecular design to the macroscopic scale of climate modeling and bibliometric analysis, GFMs are demonstrating unprecedented capabilities in representation learning, reasoning, and generation.

However, the path forward is not without obstacles. The lack of universal scaling laws, the challenge of extreme heterogeneity, and the need for interpretability and security require sustained research effort. Future advancements will likely depend on the convergence of multiple disciplines, combining insights from network science, physics, cognitive science, and computer science to build models that are not only powerful but also trustworthy and aligned with scientific principles. As the field matures, we anticipate a shift from isolated task-specific models to integrated, autonomous scientific agents capable of driving discovery in a closed-loop manner, fundamentally altering the landscape of human knowledge creation.

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